

A Tour of Obscure Summer Globular Clusters

Any list of favorite summer deep-sky objects is certain to include globular clusters such as M4, M5, M13, M15, and M22. However, adventurous observers willing to dig deeper will find more exotic quarry: huge, distant globulars residing in our galaxy's outer halo, highly reddened clusters hiding behind a veil of intervening dust in the galactic plane, and globulars that might belong to nearby dwarf galaxies.

Investigations by Harlow Shapley in 1916–1919 led to the discovery that the globulars form a spherical halo centered on the galactic hub, which lies in the direction of Sagittarius. Indeed, of the 150 known globulars, 50 percent reside in the constellations of Ophiuchus, Scorpius, and Sagittarius. (A complete list of globulars is maintained by Brian Skiff at

<ftp://ftp.lowell.edu/pub/bas/deepsky/globulars.tab>.)

Let's begin our exploration with **IC 1257**, an overlooked globular tucked away in a dusty region of Ophiuchus. Although Rudolph Spitaler discovered it more than a century ago, its true nature was unknown until recently. In 1930 Harlow Shapley classified it as an open cluster, and Sven Cederblad included it in a list of galactic nebulae in 1946. It wasn't until 1996 that a color-magnitude diagram finally revealed that IC 1257 is a distant, heavily reddened globular residing on the opposite side of the galaxy more than 80,000 light-years away. In July of 1995 I took a look with my 17½-inch f/4.5 and found a faint, round patch about 1¢ in diameter. IC 1257 looked like a 14th-magnitude galaxy with no hint of

resolution or a central concentration.

Next we drop down 32° of declination into the tail of Scorpius. **NGC 6380** is buried in a rich star field close to the galactic plane, where intervening dust clouds dim the cluster's light by four magnitudes. Until recently this was one of a few globulars whose distance was completely unknown. In 1997 it was discovered that it resides in the galactic bulge, approximately 35,000 light-years away. At –39° declination, midnorthern observers must work through haze and poor seeing to ferret out this challenging globular. To improve your chances, time your observation for when the cluster transits the meridian.

Observing from the Cape of Good Hope in 1836, John Herschel described NGC 6380 as “a star of 9th magnitude, with an extremely faint nebulous wisp or tail, extending northwards about one arcminute.” From a dark site in the Sierra Nevada foothills of California (+38° latitude), my 13-inch f/4.5 Dobsonian revealed only a low-surface-brightness haze just north of a 10th-magnitude star. Be forewarned that a number of sources list the wrong position. NGC 6380 is plotted too far east in *Uranometria 2000.0*. Search for the cluster 36¢ southwest of 4th-magnitude Q Scorpii.

You should have no problem identifying the field of **NGC 6717** — just point your scope at 5th-magnitude Nu² (ν²) Sagittarii, 4.5° east-northeast of the spectacular globular M22. With a 4-inch or larger scope, you may notice a small, faint glow just south of the glare from the star. Observing in the Sierra Nevada range at 2,200 meters (7,200 feet) elevation I was able to make out a very small bright core, and as the seeing steadied, this was re-



The Milky Way's globular clusters swarm about it like bees around a hive. The center of this hive lies in the direction of Sagittarius, and for this reason we find about half of the galaxy's globular clusters in the constellations Sagittarius, Scorpius, and Ophiuchus.

Obscure Summer Globulars

Object	Constellation	R.A. (2000.0)	Dec.	Mag. (V)	Chart number	
					<i>Uranometria</i> 2000.0	<i>Millennium</i> Star Atlas
IC 1257	Ophiuchus	17 ^h 27.1 ^m	− 7 06♠	13.1	293 ¹	1322
NGC 6380	Scorpius	17 ^h 34.5 ^m	− 39 04♠	11.5	408 ²	1459
NGC 6717	Sagittarius	18 ^h 55.1 ^m	− 22 42♠	8.4	340	1390
NGC 6749	Aquila	19 ^h 05.3 ^m	+ 1 54♠	12.4	251	1293
Terzan 7	Sagittarius	19 ^h 17.7 ^m	− 34 40♠	12.0	379	1433
Pal 10	Sagitta	19 ^h 18.0 ^m	+ 18 34♠	13.2	161	1221
Arp 2	Sagittarius	19 ^h 28.7 ^m	− 30 21♠	13.0	379	1411

¹ Not plotted on chart. ² Position plotted incorrectly.

solved into a close pair of 12.5-magnitude stars. A tight pair of 13.5-magnitude stars was also visible at the northeast edge, 25″ from the cluster's center. The 1♠ glow of NGC 6717 has a low surface brightness and appeared to me to be much smaller than the cataloged dimensions. George Abell seems to have missed the earlier NGC listing and cataloged the globular again in the 1950s as part of the Palomar Observatory Sky Survey (POSS). Because of this, you may also find NGC 6717 occasionally identified as Palomar 9.

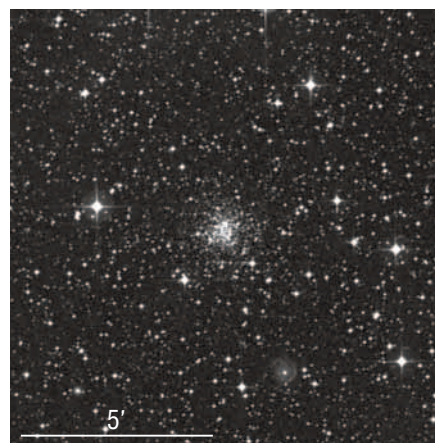
Our next targets are heavily reddened, as they lie within the Great Rift in the Aquila/Sagitta Milky Way, just a couple of degrees from the galactic plane. Obscuration by dust (of at least 4.5 magnitudes) and sparse membership combine to make **NGC 6749** one of the most challenging NGC globulars. As with IC 1257, this object has been little studied until recently and has even appeared in lists of open clusters as Berkeley 42. Because so little central brightening was visible in my 13.1-inch f/4.5 reflector, NGC 6749 looked more like a diffuse nebula set against the rich Aquila Milky Way than a globular. A few faint foreground stars are superimposed over the cluster. A poor position given for Berkeley 42 has resulted in NGC

6749 being misplotted in popular star atlases 7♠ south of its actual position. Our table, above, gives the correct position.

In the 1950s, the tally of globular star clusters increased by 13 when new globulars were discovered in the POSS. One of the more challenging is **Palomar 10**. The cluster is relatively nearby (20,000 light-years distant) but is embedded in the Sagitta Milky Way and loses five magnitudes to visual extinction. Despite an integrated visible-light-magnitude of 13.2, even detecting this dim glow, scarcely brighter than the Milky Way background, is an accomplishment. Even with my 17.5-inch under the 6.7-magnitude skies in the Sierras, I can barely glimpse it just southwest of a 13.5-magnitude star.

Globular clusters formed early in the history of the Milky Way, yet **Arp 2** and **Terzan 7** are a pair of enigmatically young globulars residing in the galactic halo. One explanation for this apparent paradox is that they are actually extragalactic in origin. They lie at the same distance as the Sagittarius Dwarf Spheroidal Galaxy, the Milky Way's nearest neighbor. This companion galaxy was discovered serendipitously in 1994, and recent studies have shown that these two, and two other globulars (M54 and Terzan 8), may have been spawned by the dwarf system. It appears likely that the dwarf and its retinue of globulars are in the process of merging with the Milky Way.

Arp 2, discovered in 1965, resides in eastern Sagittarius just 2.6 west-northwest of the globular M55. Arp 2 is a very loose cluster — search for



DIGITIZED SKY SURVEY

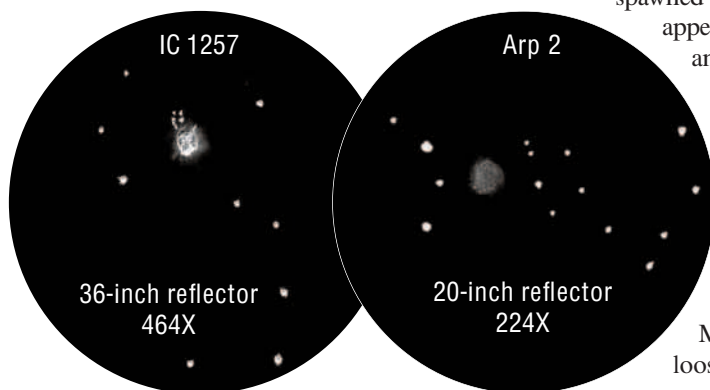
Terzan 7 (above) is one of four globular clusters believed to be members of the **Sagittarius Dwarf Spheroidal Galaxy**. The galaxy itself was discovered in 1994 and is the Milky Way's nearest neighbor, lying only 50,000 light-years from our galaxy's nucleus. The dwarf's brightest globular is M54.

a low-surface-brightness glow within a rich Milky Way field. Occasionally I could glimpse a faint star or knot at its center in my 17.5-inch scope.

In the late 1960s, the French astronomer Agop Terzan discovered 11 faint globulars. The brightest of these difficult objects is **Terzan 7**, a sparse cluster that formed some four billion years later than typical Milky Way globulars. Using my 17.5-inch at 220×, I picked up a very faint glow spanning 1.5♠. At 280×, two magnitude-14 stars were seen at the eastern edge and a 15th-magnitude star was symmetrically placed at the west-northwest edge. The cluster appeared irregularly illuminated with a few stars flickering in and out of visibility.

Successful observation of many of these clusters will require a dark site with an unobstructed southern horizon, at least a 10-inch telescope, and lots of patience! Although most are plotted in *Uranometria 2000.0*, you'll find a more detailed computer-generated finder chart or the *Millennium Star Atlas* essential aids to pinpointing their exact positions. Experiment with a variety of magnifications and be prepared to hone your observing skills as you search for these challenging summer globulars.

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Ace visual observer Barbara Wilson produced these sketches of two of the sky's lesser-known globulars. South is up in both drawings.